LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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SET 21: Energy Consumption Analysis

Aim

To analyze energy consumption using a histogram, density plot, bubble scatter plot, sector-wise renewable energy bar chart, and an interactive Tableau dashboard.

Dataset: Energy_Consumption_Data.csv

Use this structured version for the first five clearly identifiable energy records, plus the final South-region March record:

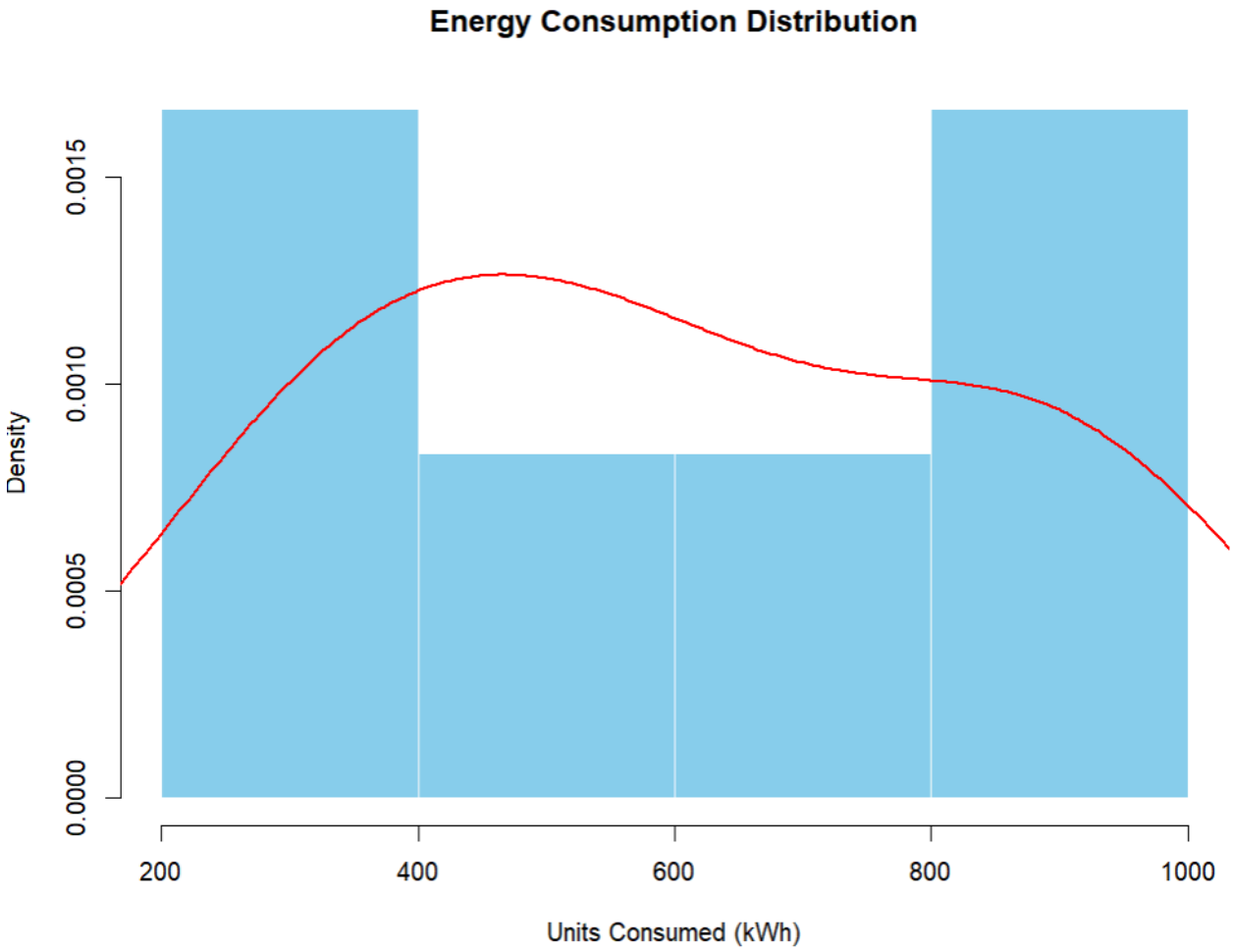
Region	Month	Temperature (°C)	Units_Consumed (kWh)	Cost (₹)	Renewable_Usage (%)	Sector	Peak_Hours
North	Jan	15	320	2100	22	Residential	4
South*	Jan	24	540	3600	18	Commercial	6
West	Feb	20	880	5900	12	Industrial	8
East	Feb	18	350	2300	25	Residential	5
North*	Mar	28	610	4100	20	Commercial	7
South	Mar	30	920	6200	15	Industrial	9

Q1. Plot a histogram and density plot for Units_Consumed

R Program

```
data <- read.csv("Energy_Consumption_Data.csv")
# Histogram
hist(
  data
Units_Consumed, probability = TRUE, main = "Energy Consumption Distribution", xlab =
Units_Consumed)
lines(density_curve, col = "red", lwd = 2)
```

Output:



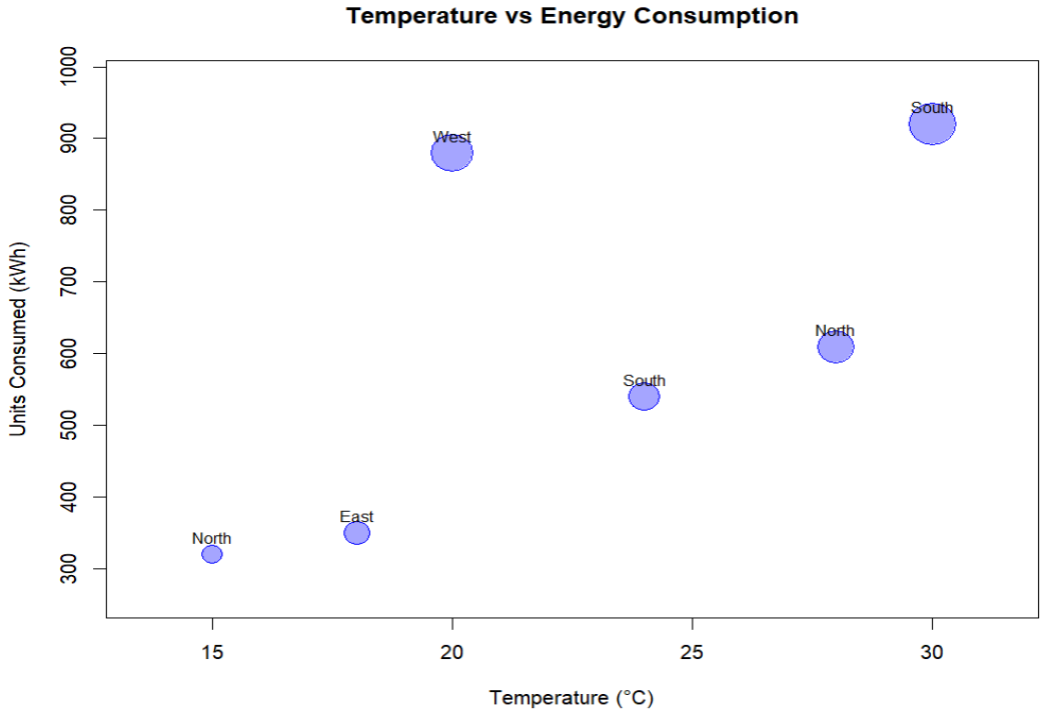
Observation: The values range from 320 to 920 kWh. The density curve helps show where consumption values are concentrated. Since the dataset contains only six observations, the distribution should be interpreted cautiously.

Q2. Create a bubble scatter plot between Temperature and Units Consumed

R Program

```
data <- read.csv("Energy_Consumption_Data.csv")
symbols(
  x = data$Temperature, y = data$Units_Consumed,
  circles = data$Peak_Hours/max(data$Peak_Hours) * 0.5,
  inches = FALSE,
  bg = rgb(0, 0, 1, alpha = 0.35),
  fg = "blue",
  xlab = "Temperature (°C)",
  ylab = "Units Consumed (kWh)",
  main = "Temperature vs Energy Consumption"
)
text(
  data$Temperature, data$Units_Consumed,
  labels = data$Region,
  pos = 3,
  cex = 0.8
)
```

Output:



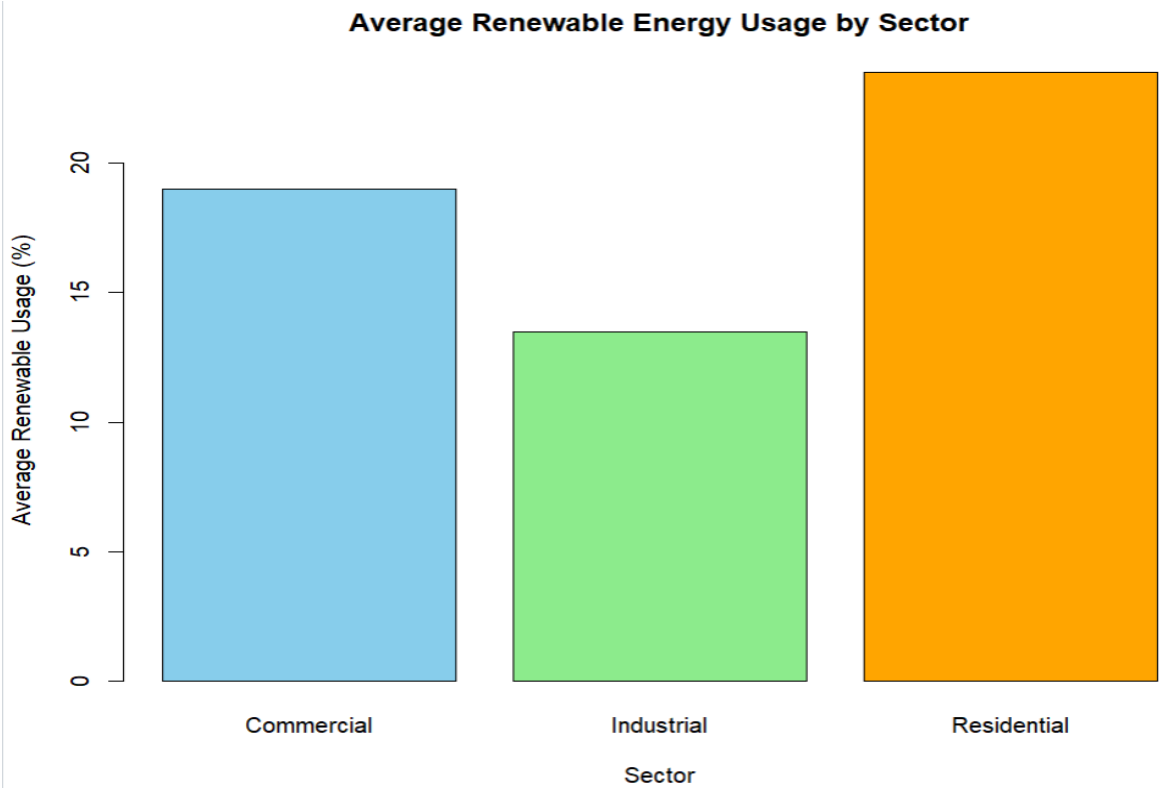
Observation: The higher-temperature observations have relatively high consumption in this small dataset. More observations are needed to establish a reliable relationship.

Q3. Calculate average renewable usage for each sector and create a bar chart

R Program

```
data <- read.csv("Energy_Consumption_Data.csv")
average_renewable <- aggregate(
  Renewable_Usage ~ Sector,
  data = data,
  FUN = mean
)
print(average_renewable)
barplot(
  average_renewableRenewable_Usage, names.arg = average_renewable
Sector,
  xlab = "Sector",
  ylab = "Average Renewable Usage (%)",
  main = "Average Renewable Energy Usage by Sector",
  col = c("skyblue", "lightgreen", "orange")
)
```

Output:



Observation: Residential has the highest average renewable usage at 23.5%, while Industrial has the lowest at 13.5%.

Q4. Create an interactive Tableau dashboard

Step 1 — Prepare the CSV file

Save the following content as Energy_Consumption_Data.csv.

```
Region,Month,Temperature,Units_Consumed,Cost,Renewable_Usage,Sector,Peak_Hours
North,Jan,15,320,2100,22,Residential,4
South,Jan,24,540,3600,18,Commercial,6
West,Feb,20,880,5900,12,Industrial,8
East,Feb,18,350,2300,25,Residential,5
North,Mar,28,610,4100,20,Commercial,7
South,Mar,30,920,6200,15,Industrial,9
```

These region and sector assignments use the assumptions explained above.

Step 2 — Import the dataset

1. Open Tableau.
2. Select Connect → Text File.
3. Select Energy_Consumption_Data.csv.
4. Check that the fields are recognized correctly.

Step 3 — Create the monthly energy usage line chart

1. Create a new worksheet named Monthly Energy Usage.
2. Drag Month → Columns.
3. Drag Units_Consumed → Rows.
4. Change the Marks dropdown to Line.
5. Drag Region → Color to compare regions, or use AVG(Units_Consumed) for average monthly consumption.
6. Set the title to Monthly Energy Consumption Trends.

If Tableau sorts months alphabetically, manually sort them as Jan, Feb, Mar.

Step 4 — Create the heatmap

1. Create a new worksheet named Energy Consumption Heatmap.
2. Drag Region → Columns.
3. Drag Sector → Rows.
4. Change the Marks dropdown to Square.
5. Drag Units_Consumed → Color.
6. Drag Units_Consumed → Label.
7. Set the title to Energy Consumption by Sector and Region.

Step 5 — Add filters

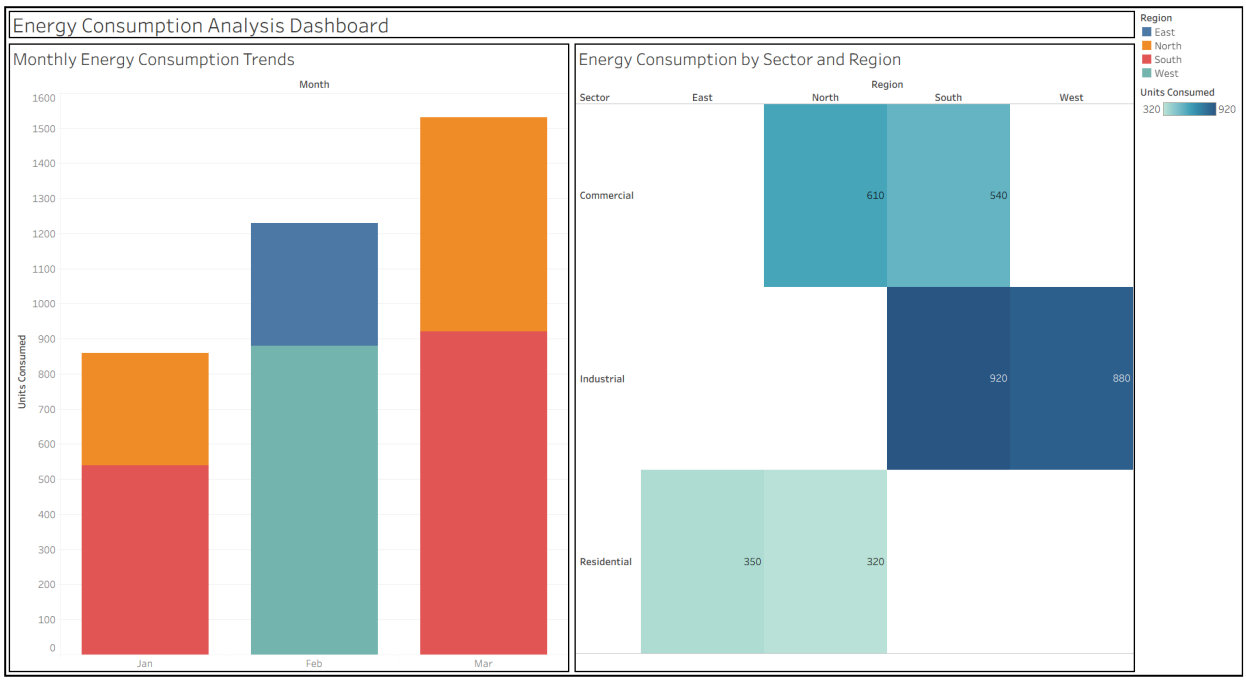
1. Drag Month → Filters and select the months you want.
2. Drag Sector → Filters.
3. Right-click each filter and choose Show Filter.

Step 6 — Create the dashboard

1. Click New Dashboard.
2. Add the Monthly Energy Usage worksheet.

3. Add the Energy Consumption Heatmap worksheet.
4. Show the Month and Sector filters.
5. Arrange the charts neatly and set the dashboard title to Energy Consumption Analysis Dashboard.

Dashboard:



Procedure

1. Imported the energy consumption dataset into R.
2. Created a histogram with a density curve.
3. Generated a bubble scatter plot using temperature, consumption, and peak hours.
4. Calculated average renewable usage by sector and created a bar chart.
5. Imported the CSV file into Tableau and created a monthly line chart and heatmap.
6. Combined the visualizations and filters into an interactive dashboard.

Result

Energy consumption patterns, temperature relationships, and sector-wise renewable energy usage were analyzed using R. A Tableau dashboard combining monthly energy trends, a sector-region heatmap, and interactive filters was designed successfully.